MILITARY SPECIFICATION SHEET

GENERATOR SET, GASOLINE ENGINE DRIVEN, 0.5 KW, 28 VOLT DIRECT CURRENT, TYPE I (TACTICAL), CLASS 2 (UTILITY)

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the generator set described herein shall consist of this document and the issue in effect of specification MIL-G-52732.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for mode IV, size 0.5 sets, model MEP-024A.

1.2 Classification.

1.2.1 Set rating. The mode IV set shall be rated 28 volts direct current (VDC).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-G-52732	- Generator Sets, Gasoline Engine Driven, 0.5 KW thru 10 KW, 60 Hertz, 400 Hertz and 28 Volt Direct Current, Type I (Tactical), Class 2 (Utility) General Specification for

STANDARDS

Military

MS39297	- Engine, Gasoline, Air-Cooled, Industrial-Type, 4 Cycle, 1-1/2 Net Continuous HP.

FSC 6115

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DRAWINGS

ME

TA13217E1750 - Generator Set, Gasoline Engine Driven, 0.5 KW, 28 V DC.

(Copies of specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Requirements shall be in accordance with MIL-G-52732, and as specified herein.

3.2 Description. The set shall be in accordance with Top Assembly TA13217E1750 and as specified herein.

3.3 Starting and operating. Servicing, adjustment, and replacement shall be as specified in table I.

3.3.1 With winterization. Not applicable for the mode IV, size 0.5 set.

3.4 Maximum power. The maximum power (corrected) in kilowatts of the set shall be not less than 0.65 kw when operated at rated speed, voltage, and at unit power factor.

3.5 Engine.

3.5.1 Engine design. The Government furnished engine will conform to MS39297.

3.5.2 Engine temperature. Engine temperatures shall not exceed the following:

Location	Maximum Temperature	
	First 10 hours of engine operation	After 10 hours of engine operation
Sump oil	300° F. (148.9° C.)	250° F. (121° C.)
Cylinder head	500° F. (260° C.)	475° F. (246.1° C.)

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Table I. Schedule of Servicing and Adjusting

All Items May Be Serviced Before Test

Item No.	Hours between service periods	Item
1	8	Any minor maintenance such as retightening and retorquing nuts, bolts and screws that can be performed without disturbing or removing adjacent components, accessories or shrouding (except top shroud pump, intake manifold, and exhaust system).
2	Not allowed	Any major maintenance that requires removal of other components, accessories or shrouding such as oil pans, cylinder heads and connecting rods.
3	As required	Cleaning and regapping or replacing spark plug.
4	As required	Servicing fuel filter.
5	100	Checking compression pressure.
6	4	Checking and adding lubricating oil, as necessary.
7	8	Adjusting carburetor.
8	As required	Cleaning carburetor.
9	8	Adjusting governor (normal speed adjustment).
10	50	Changing lubricating oil.
11	100	Servicing air cleaner.
12	50	Adjusting and dressing breaker points.
13	50	Adjusting ignition timing.

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Table I. Schedule of Servicing and Adjusting (continued)

All Items May Be Serviced Before Test

Item No.	Hours between service periods	Item
14	250	Valve adjustment.
15	1500	Cleaning and inspection of cumbustion chamber, manifolds, and valves.
16	250	Replacing breaker points.
17	As required	Replacing condenser.
18	*	Gasket replacement.

* Not allowed except where component is removed for servicing.

3.5.3 Governing system.

3.5.3.1 Short term, steady-state stability (30 seconds), mode II and III sets. Not applicable for the mode IV, size 0.5 set.

3.5.3.2 Long term, steady-state stability (4 hours), mode II and III sets. Not applicable for the mode IV, size 0.5 set.

3.5.3.3 Transient performance, mode II and III sets. Not applicable for the mode IV, size 0.5 set.

3.6 Generator.

3.6.1 Voltage unbalance, mode II and III sets. Not applicable for the mode IV, size 0.5 set.

3.6.2 Phase balance (voltage), mode II and III sets. Not applicable for the mode IV, size 0.5 set.

3.6.3 Voltage waveform, mode II and III sets. Not applicable for the mode IV, size 0.5 set.

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3.6.4 Output voltage ripple, mode IV sets. The output voltage ripple shall not exceed 5.5 percent of rated voltage.

3.6.5 Saturation curves. With the generator temperature stabilized at normal ambient conditions the exciter field current for rated voltage shall not exceed 0.25 ampere at no load and 0.40 ampere at rated load.

3.6.6 Voltage adjustment range. The minimum acceptable voltage adjustment range at normal ambient conditions shall be 23 to 35 volts. The minimum voltage adjustment range at extreme temperature conditions of minus 25° F. (minus 31.7° C.) and plus 125° F. (51.7° C.) shall be 5 percent above and below rated voltage.

3.6.7 Voltage regulation. The voltage regulation percentage shall not exceed 4 percent.

3.6.8 Voltage steady-state performance. The voltage steady-state stability bandwidth percentage shall not exceed 2 percent.

3.6.9 Voltage transient performance. The voltage dip percentage shall not exceed 30 percent and the voltage rise percentage shall not exceed 40 percent.

3.6.10 Phase sequence (rotation), mode II and III sets. Not applicable for the mode IV, size 0.5 set.

3.6.11 Generator efficiency. The generator efficiency shall not be less than 68 percent.

3.7 Circuit interrupter. The circuit interrupter shall open between 0.85 and 4.0 seconds with 200 percent load applied.

3.8 Instruments. The inaccuracy of the metering system shall not exceed the following:

Instrument	Allowable Inaccuracy
Ammeter (percent load)	10 percent
Voltmeter	3 percent

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4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance provisions. The quality assurance provisions shall be in accordance with MIL-G-52732 and as specified herein.

4.2 Engine preventative maintenance services. The engine preventative services shall be performed as specified in table I.

4.3 Drop test (free fall). The height of the drop shall be 18 inches for the mode IV, size 0.5 set.

4.4 Vibration test. The accelerometer used to compute transmissibility shall be located at the top of the engine adapter plate on the mode IV, size 0.5 set. The resonant dwell portion of the test is not required on the mode IV, size 0.5 set.

4.5 Railroad impact test. The railroad impact test shall be performed with the mode IV, size 0.5 set packaged level A.

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery. Preparation for delivery shall be in accordance with MIL-G-52732.

6. NOTES

6.1 Notes. The notes specified in MIL-G-52732 are applicable to this specification and as specified herein.

6.2 Stable operation (voltage). Stable operation is operation within a bandwidth of 2 percent of the rated voltage.

6.3 International standardization. Not applicable for the mode IV, size 0.5 set.

Custodians:

Army - ME
Air Force - 80
Navy - YD

Review activities:

Army - EL, MI, MU

Preparing activity:

Army - ME

Project No. 6115-0144-3

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION: MIL-G-52732, 3 - Generator Set, Gasoline Engine Driven, 0.5 KW 28 Volt Direct Current, Type I (Tactical), Class 2 (Utility)

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO (If "yes" in what way?)

4. REMARKS (Attach any pertinent data which may be of use in improving this specification. If there are additional papers attach to form and place both in an envelope addressed to preparing activity.)

SUBMITTED BY (Printed or typed name and activity - Optional) DATE

DD FORM 1426

REPLACES EDITION OF 1 OCT 54 WHICH MAY BE USED.

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U. A. ARMY MOBILITY EQUIPMENT COMMAND
RESEARCH, DEVELOPMENT AND ENGINEERING DIRECTORATE
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FORT BELVOIR, VIRGINIA 22060

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